

Booleans

In Python, a Boolean always has a value that is either “True” or “False”. In both cases, the word begins with a capital letter.

▷ True

When the value “True” is put into a variable, it will be a Boolean variable.

```
>>> a = True
>>> print(a)
True
```

No quote marks

Boolean value printed

▷ False

When the value “False” is put into a variable, it will be a Boolean variable too.

```
>>> a = False
>>> print(a)
False
```

Boolean value printed



EXPERT TIPS

Spotting data types

In Python, there are many data types. To find out what data type something is, you can use the “type” command.

```
>>> type(24)
<class 'int'>
>>> type(24.3)
<class 'float'>
>>> type('24')
<class 'str'>
```

“type” command

24 is an integer (“int”)

24.3 is a float (“float”)

‘24’ is a string (“str”) because it is in quote marks

Converting data types

Variables can contain any type of data. Problems occur if you try to mix types together. Data types sometimes have to be converted; otherwise, an error message will appear.



▷ Mixed type

The “input” command always gives a string, even if a number is entered. In this example, since “apple” actually contains a string, an error message is displayed.

```
>>> apple = input('Enter number of apples ')
Enter number of apples 2
>>> print(apple + 1)
TypeError
```

Variable name

String in quote marks shown on screen

Tries to add the number 1 to the variable “apple”

The program gives an error message because Python doesn't know how to add a number to a string

▷ Converting data types

To convert the string into a number, the “int()” command is used to turn it into an integer.

```
>>> print(int(apple) + 1)
3
```

The program now works and shows the result

The variable turns from a string into an integer, so a number can be added to it